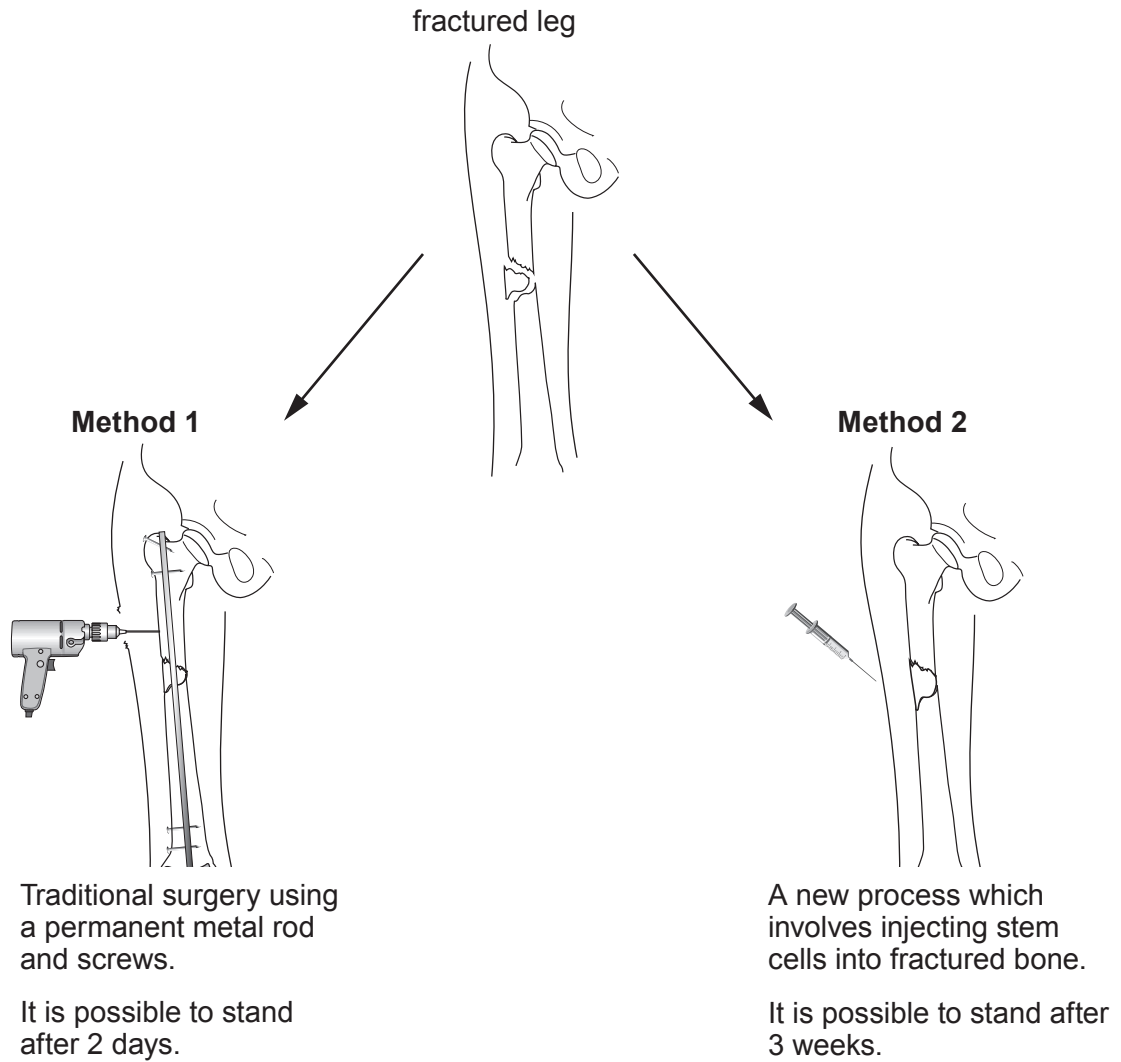


Answer all questions

1. The photograph below shows a young ice skater, Tracey, who falls and fractures her leg.



The diagram below shows two ways in which Tracey’s fractured leg could be treated in a large modern hospital.



Expected results of treatment

Time after treatment (weeks)	Percentage of bone healing (%)	
	Traditional surgery	Injection of stem cells
10	10	12
20	14	25
30	19	38
40	28	55
50	41	70
60	59	82

- (a) Give **one** reason why Tracey might choose traditional surgery and **one** reason why, alternatively, she might prefer to be treated by an injection of stem cells. [2]

.....

.....

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- (b) (i) When fractured bones heal, cells divide and multiply by mitosis.

Complete the table below.

[2]

Mitosis in human cells	
chromosomes in mother cell	46
number of daughter cells produced after one division	
number of chromosomes in daughter cells	

- (ii) Following mitosis, what must happen to stem cells, in order for them to repair the fractured bone? [1]

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(iii) State the name of the disease that can occur if cell division by mitosis is uncontrolled. [1]

.....

(c) Stem cells can be obtained from both adults and embryos.

Give **one** reason why some people have a strong personal objection to the use of embryonic stem cells in medical research. [1]

.....

7

